RADIOACTIVITY

ICSE 10

CHAPTER NOTES

Section 1: Introduction to Radioactivity

Definition of Radioactivity

Radioactivity is the spontaneous disintegration or decay of unstable atomic nuclei, releasing radiation in the form of alpha, beta, or gamma rays.

Cause of Radioactivity

Unstable nuclei undergo radioactive decay to achieve stability. This occurs due to an imbalance in the ratio of protons to neutrons in the nucleus.

Natural and Artificial Radioactivity

1. **Natural Radioactivity**: Occurs naturally in elements like uranium, radium, and thorium.
2. **Artificial Radioactivity**: Induced in stable elements by bombarding them with neutrons or other particles (like in nuclear reactors).

Discovery of Radioactivity

Henri Becquerel discovered radioactivity in 1896 when he observed that uranium salts emitted invisible radiation that affected photographic plates.

Section 2: Types of Radiation

1. Alpha (α) Radiation

- Nature: Helium nuclei (2 protons and 2 neutrons)
- Charge: Positive (+2)
- Mass: Heavy (mass = 4 atomic mass units or amu)
- Penetration: Low (can be stopped by paper or skin)
- Ionizing Ability: High (due to heavy mass and charge)

2. Beta (β) Radiation

- Nature: High-speed electrons (β^-) or positrons (β^+)
- Charge: Negative (β^-) or Positive (β^+)
- Mass: Very small (1/1836 of a proton)
- Penetration: Medium (can be stopped by aluminum foil)
- Ionizing Ability: Moderate

3. Gamma (γ) Radiation

- Nature: Electromagnetic waves (photons) with no mass or charge
- Charge: Neutral (0)
- Mass: None (pure energy)
- Penetration: Very high (can pass through thick lead and concrete)
- Ionizing Ability: Low but harmful due to high penetration

Section 3: Radioactive Decay

Definition of Decay
Radioactive decay is the process by which an unstable nucleus transforms into a more stable nucleus by emitting radiation.

Types of Decay

1. **Alpha Decay**:
   - The nucleus emits an alpha particle (2 protons, 2 neutrons), reducing its mass number by 4 and its atomic number by 2.
   - Example:
     Uranium-238 decays to Thorium-234:
     $U_{92}^{238} \rightarrow Th_{90}^{234} + He_{2}^{4}$
2. **Beta Decay**:
   - A neutron in the nucleus changes into a proton, releasing a beta particle (electron) and an antineutrino.
     Carbon-14 decays to Nitrogen-14:
     $C_{6}^{14} \rightarrow N_{7}^{14} + e_{-1}^{0}$
3. **Gamma Decay**:
   - The nucleus releases excess energy as gamma rays after alpha or beta decay

Section 4: Half-Life and Decay Constant

Half-Life ($T_{½}$)
The half-life of a radioactive substance is the time required for half of the radioactive nuclei in a sample to decay.

Formula
$N = N_0 (1/2)^n$

Where,
N = number of undecayed nuclei remaining
N_0 = initial number of nuclei
n = number of half-lives elapsed

Decay Constant (λ)
The decay constant (λ) is the probability of decay of a nucleus per unit time.

Relationship Between Half-Life and Decay Constant
$T^{½} = 0.693 / \lambda$

Section 5: Mass-Energy Equivalence ($E = mc^2$)

Einstein's Mass-Energy Relation
The equation **$E = mc^2$** shows the relationship between mass and energy, where a small amount of mass (m) can be converted into a large amount of energy (E).

Formula
$E = mc^2$

Where,
E = energy (Joules)
m = mass (kg)
c = speed of light (3×10^8 m/s)

Applications

1. Nuclear fission and fusion release energy due to mass loss (mass defect).
2. Energy produced in nuclear reactors and atomic bombs is based on this principle.

Section 6: Units Used in Radioactivity (MeV, amu, Curie, Becquerel)

1. Mega Electron Volt (MeV)

- 1 MeV = 1.6×10^{-13} Joules
- It is a unit of energy commonly used in nuclear physics.
- Example: Energy released during the decay of radioactive elements is often expressed in MeV.

2. Atomic Mass Unit (amu)

- 1 amu = 1/12th the mass of a carbon-12 atom.
- 1 amu = 1.66×10^{-27} kg.
- It is used to express the masses of atoms and subatomic particles.

3. Curie (Ci) and Becquerel (Bq)

- 1 Becquerel (Bq) = 1 decay per second (SI unit of activity).
- 1 Curie (Ci) = 3.7×10^{10} decays per second (old unit).

Section 7: Nuclear Fission and Nuclear Fusion

Nuclear Fission

- In nuclear fission, a heavy nucleus (like Uranium-235) splits into two smaller nuclei, releasing energy.
- **Example**:
  $U_{92}^{235} + n \rightarrow Ba_{56}^{144} + Kr_{36}^{89} + 3n + Energy$
- Energy released in nuclear fission is due to the mass defect and is calculated using $\mathbf{E = mc^2}$.
- **Applications**:
  - Used in nuclear power plants to generate electricity.
  - Used in atomic bombs

Nuclear Fusion

- In nuclear fusion, two light nuclei (like hydrogen) combine to form a heavier nucleus, releasing energy.
  $H_1^2 + H_1^3 \rightarrow He_2^4 + n_0^1 + Energy$
- Fusion releases much more energy than fission but requires extremely high temperatures.

- **Applications**: Occurs naturally in stars like the sun, where hydrogen nuclei fuse to form helium. Research is being conducted to create fusion reactors for energy production.

Section 8: Detection of Radiation

Devices to Detect Radiation

1. **Geiger-Muller Counter**: Detects alpha, beta, and gamma radiation using ionization of gas inside a tube.
2. **Cloud Chamber**: Detects alpha and beta particles by observing the condensation trails of vapor.
3. **Photographic Film**: Used in radiation badges worn by workers exposed to radiation.
4. **Scintillation Counter**: Uses a phosphor-coated material to detect and count radiation events.

Section 9: Applications of Radioactivity

Medical Applications

- **Cancer Treatment**: Gamma rays are used in radiotherapy to kill cancer cells.
- **Radioactive Tracers**: Isotopes like Iodine-131 are used to detect thyroid gland disorders.

Industrial Applications

- **Thickness Control**: Beta radiation controls the thickness of materials like paper and metal sheets.
- **Leak Detection**: Radioactive tracers detect leaks in pipelines.

Scientific Research

- Radiocarbon dating (C-14) is used to estimate the age of fossils.

Energy Generation

- Nuclear reactors generate power using nuclear fission of Uranium-235.

Section 10: Summary of Key Formulas and Definitions

1. **Mass-Energy Relation**:
   $E = mc^2$
2. **Half-Life**:
   $N = N_0(1/2)^n$
3. **Decay Constant**:
   $T½ = 0.693 / \lambda$
4. **Radioactive Decay**:
   Alpha Decay:
   $U_{92}^{238} \rightarrow Th_{90}^{234} + He_2^4$
5. **Nuclear Fission**:
   $U_{92}^{235} + n \rightarrow Ba_{56}^{144} + Kr_{36}^{89} + 3n + Energy$

1. Atomic Mass Unit (amu)

- 1 amu = **1.66×10^{-27} kg**
- 1 amu = **1/12th** the mass of a carbon-12 atom.

2. Speed of Light (c)

- **$c = 3 \times 10^8$ m/s**
- This is used in the formula **$E = mc^2$** (mass-energy equivalence).

3. Avogadro's Number (N_a)

- **$N_a = 6.022 \times 10^{23}$**
- This constant is used to relate the number of atoms or molecules in a mole of a substance.

4. 1 Curie (Ci) in Becquerels

- **1 Curie (Ci) = 3.7×10^{10} decays per second (Bq)**
- A unit of radioactivity, where 1 Ci corresponds to the activity of 1 gram of radium-226.

5. 1 Becquerel (Bq)

- **1 Bq = 1 decay per second**
- SI unit of activity in radioactivity.

6. Energy of 1 Electron Volt (eV)

- **1 eV = 1.6 x 10^{-19} Joules**
- The energy gained by an electron when accelerated through a potential difference of 1 volt.

7. Energy of 1 Mega Electron Volt (MeV)

- **1 MeV = 1.6 x 10^{-13} Joules**
- Used in nuclear reactions and the energy released during radioactive decay.

8. Half-Life of Common Isotopes

- **Carbon-14 (C-14)**:
  - **Half-life = 5730 years**
  - Used in radiocarbon dating.
- **Uranium-238 (U-238)**:
  - **Half-life = 4.468 billion years**
  - Used in dating geological samples.
- **Radon-222 (Rn-222)**:
  - **Half-life = 3.8 days**
- **Iodine-131 (I-131)**:
  - **Half-life = 8.02 days**
  - Used in medical treatments and diagnostics.

9. Decay Constant (λ)

- **Decay constant (λ)** is a characteristic constant for each radioactive substance.
- Its relationship with half-life (T½) is given by the formula: $T_{1/2}=0.693\lambda$

10. Boltzmann Constant (k)

- **k = 1.38 x 10^{-23} J/K**
- Though more commonly used in thermodynamics, it is useful in understanding the statistical behavior of particles in radioactive materials at the atomic level.

11. Binding Energy Per Nucleon (in Nuclear Reactions)

- **Binding energy** is the energy required to break a nucleus into its constituent protons and neutrons.
- **Binding energy per nucleon** can be derived from mass defect using **$E = mc^2$**.

In the study of **Radioactivity**, several elements play an important role due to their radioactive properties. Below is a list of key **radioactive elements** commonly discussed in the chapter:

1. Uranium (U)

- **Isotopes**: **Uranium-238 (U-238)** and **Uranium-235 (U-235)**
- **Importance**:
  - Uranium-238 is the most abundant naturally occurring radioactive isotope and is used in radiometric dating to determine the age of rocks and fossils.
  - Uranium-235 is used in nuclear reactors and atomic bombs as it undergoes fission, releasing energy.
- **Half-life**:
  - **U-238**: 4.468 billion years (used in dating rocks).
  - **U-235**: 703.8 million years.

2. Thorium (Th)

- **Isotope**: **Thorium-232 (Th-232)**
- **Importance**:
  - Thorium is used as a potential fuel in nuclear reactors.
  - Thorium-232 is a fertile material that can be converted into Uranium-233 through neutron bombardment.
- **Half-life**:
  - **Th-232**: 14 billion years.

3. Radium (Ra)

- **Isotope**: **Radium-226 (Ra-226)**
- **Importance**:
  - Radium was once widely used in luminous paints for clocks, watches, and instrument dials due to its ability to glow in the dark.
  - It has a significant historical importance in the study of radioactivity.
- **Half-life**:
  - **Ra-226**: 1600 years.

4. Radon (Rn)

- **Isotope**: **Radon-222 (Rn-222)**
- **Importance**:
  - Radon is a noble gas that is produced from the decay of radium.
  - It is a health hazard when it accumulates in poorly ventilated spaces like basements, as it can lead to lung cancer due to its radioactive decay.
- **Half-life**:

    - **Rn-222**: 3.8 days.

5. Carbon (C)

- **Isotope**: **Carbon-14 (C-14)**
- **Importance**:
    - Carbon-14 is used in **radiocarbon dating**, a method for dating ancient organic materials.
    - It decays into nitrogen-14 by beta decay.
- **Half-life**:
    - **C-14**: 5730 years.

6. Iodine (I)

- **Isotope**: **Iodine-131 (I-131)**
- **Importance**:
    - Iodine-131 is used in **medical treatments**, especially in the treatment of thyroid cancer and hyperthyroidism.
    - It is a beta emitter and can be used as a tracer in medical imaging.
- **Half-life**:
    - **I-131**: 8.02 days.

7. Cesium (Cs)

- **Isotope**: **Cesium-137 (Cs-137)**
- **Importance**:
    - Cesium-137 is a byproduct of nuclear fission reactions and is used in industrial radiography.
    - It is also a concern in nuclear accidents due to its long half-life and potential environmental contamination.
- **Half-life**:
    - **Cs-137**: 30.17 years.

8. Polonium (Po)

- **Isotope**: **Polonium-210 (Po-210)**
- **Importance**:
    - Polonium-210 is highly toxic and is used as a heat source in space equipment.
    - It is known for its role in poisoning cases (e.g., the assassination of Alexander Litvinenko).
- **Half-life**:
    - **Po-210**: 138.4 days.

9. Plutonium (Pu)

- **Isotope**: **Plutonium-239 (Pu-239)**
- **Importance**:
  - Plutonium-239 is used as a fuel in nuclear reactors and in nuclear weapons.
  - It undergoes alpha decay and is highly toxic if inhaled or ingested.
- **Half-life**:
  - **Pu-239**: 24,100 years.

10. Neptunium (Np)

- **Isotope**: **Neptunium-239 (Np-239)**
- **Importance**:
  - Neptunium-239 is a byproduct of uranium-238 in nuclear reactors.
  - It decays into Plutonium-239, which is important in nuclear reactions.
- **Half-life**:
  - **Np-239**: 2.356 days.

11. Barium (Ba)

- **Isotope**: **Barium-140 (Ba-140)**
- **Importance**:
  - Barium-140 is produced in nuclear reactors during fission reactions.
  - It is a key component of radioactive waste and a gamma emitter.
- **Half-life**:
  - **Ba-140**: 12.8 days.

12. Krypton (Kr)

- **Isotope**: **Krypton-85 (Kr-85)**
- **Importance**:
  - Krypton-85 is produced in nuclear reactors and is used in various applications, including as a tracer in atmospheric studies.
  - It is also used in radiation therapy and some types of light sources.
- **Half-life**:
  - **Kr-85**: 10.76 years.